

Can we classify news articles as fake or real based on their textual content?

Classifying News Types from Using Machine Learning, Deep Learning and Transformer.

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Problem Statement

- **Goal:** distinguish between fake news and real news.
- **Success Criteria:** DSM + $\geq 85\%$ accuracy.
- **Scope:** supervised classification.
- **Constraints:** missing data, class imbalance, feature engineering.

Dataset Description

- WELFake_Dataset.csv.
- 72,134 entries.
- Balanced classes.
- Title, text, label features.



Available at: <https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification>

Data Wrangling

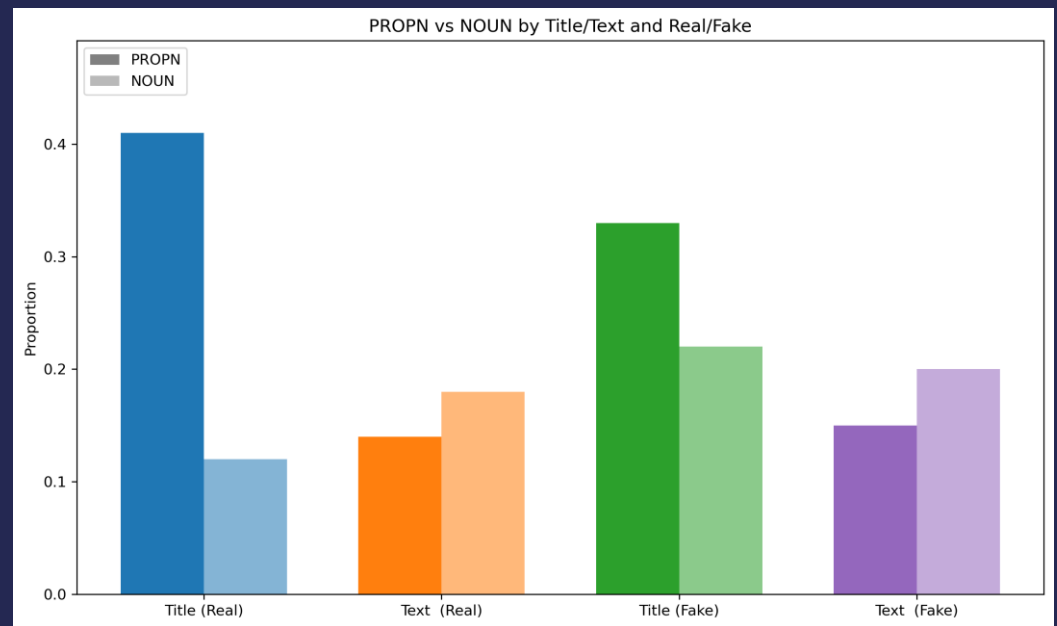
- Missing values removal (<5% nulls).
- Identification & removal of duplicates.
- Text normalization:
 - Tokenization.
 - Punctuation, digits and stop words removal.
 - Final lemmatization & character lowering.

Exploratory Data Analysis – EDA.

- **Strong** discriminators:
 - POS proper noun/other noun ratio title.
 - NER ratio title.
 - Polarity title.
 - Polarity text.
 - Subjectivity title.
 - Subjectivity text.
 - Readability title.
 - Readability text.
- **Weak** discriminators:
 - Length of text.
 - Vocabulary richness.
 - Terms frequency.
 - POS proper noun/other noun ratio text.
 - NER ratio text.
 - NER frequency.

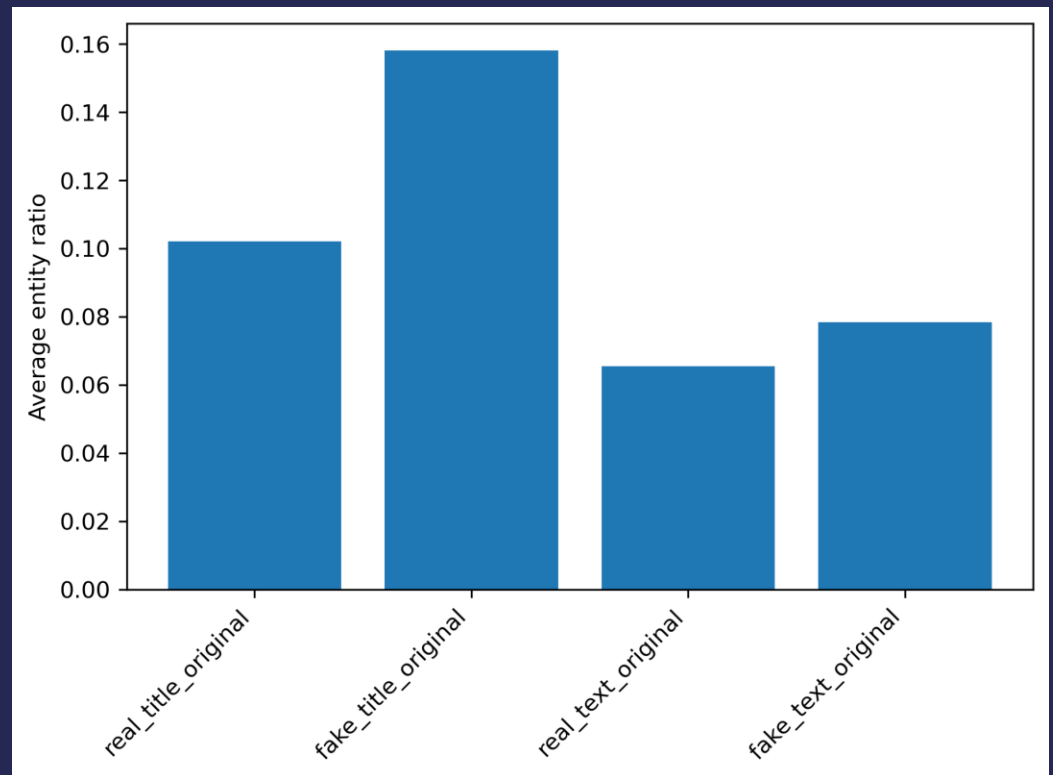
Exploratory Data Analysis – EDA.

- **POS – proper nouns & other nouns ratio:**
- Fake news title = generic nouns.
- Real news title = proper nouns.



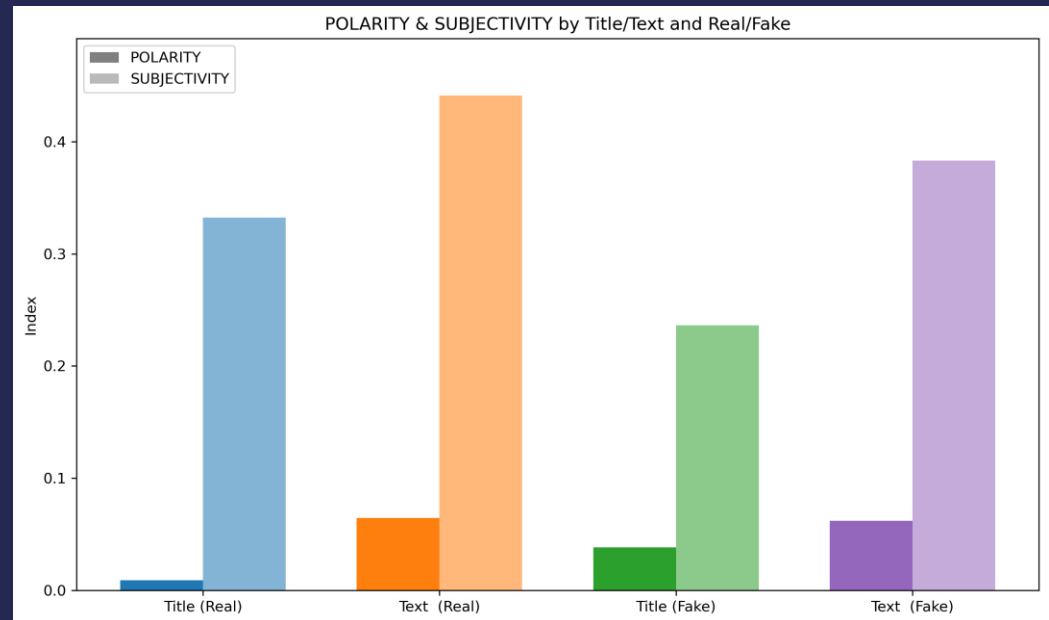
Exploratory Data Analysis – EDA.

- **NER – Named Entities Recognition:**
- Fake news title = more named entities.



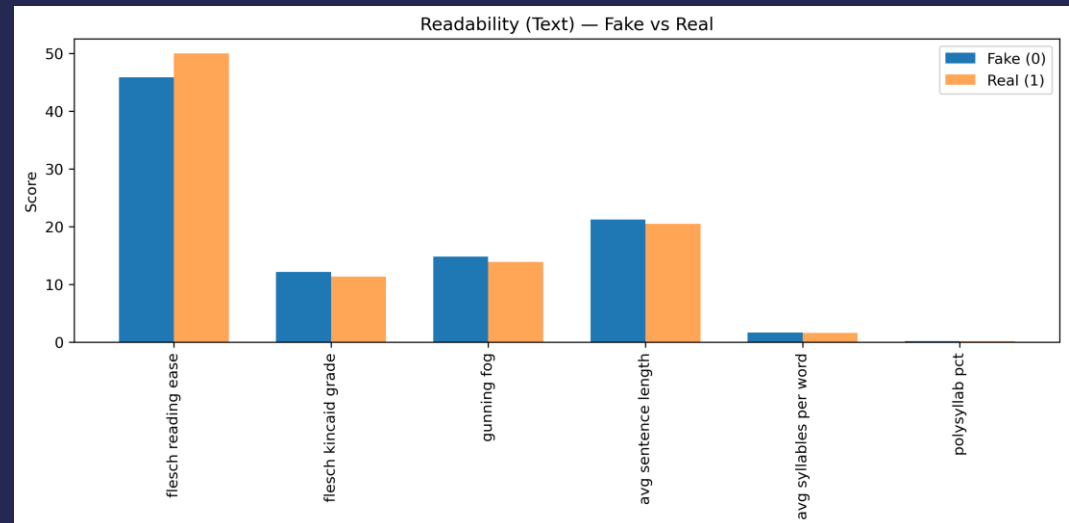
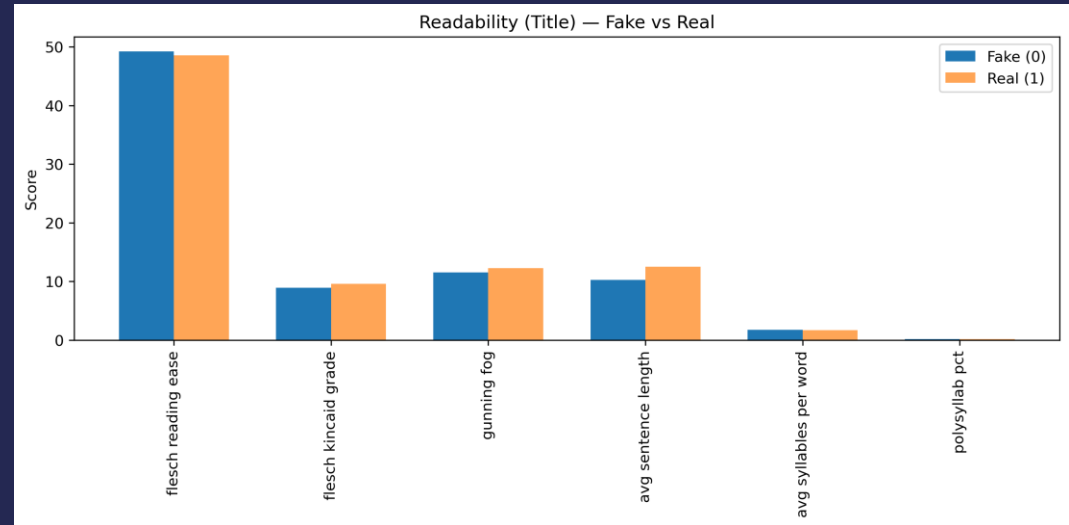
Exploratory Data Analysis – EDA.

- **Polarity & subjectivity:**
- Fake news title = **more** polarity.
- Real news title = **more** subjectivity.
- Real news text = **more** subjectivity.



Exploratory Data Analysis – EDA.

- **Readability:**
- Fake news title = **easier** to read.
- Real news text = **easier** to read.



Modeling Approach

4 models:

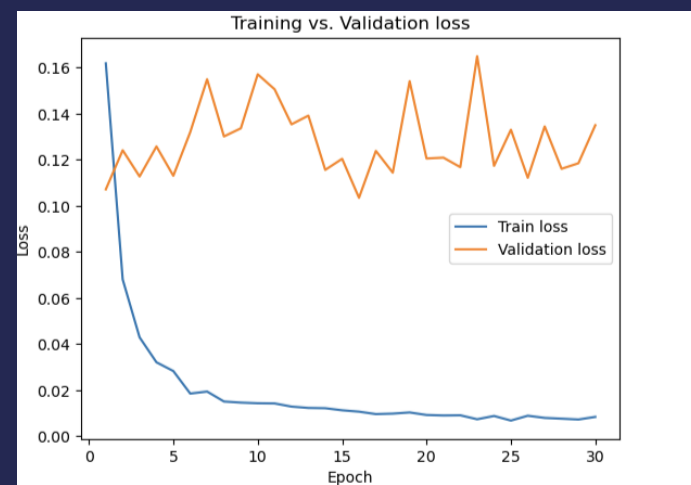
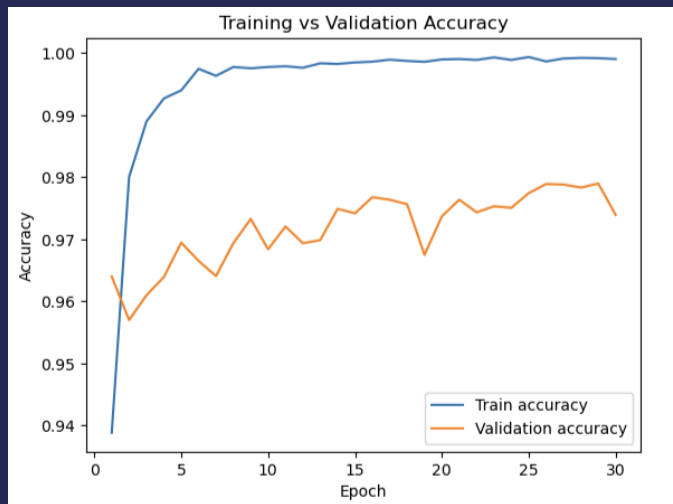
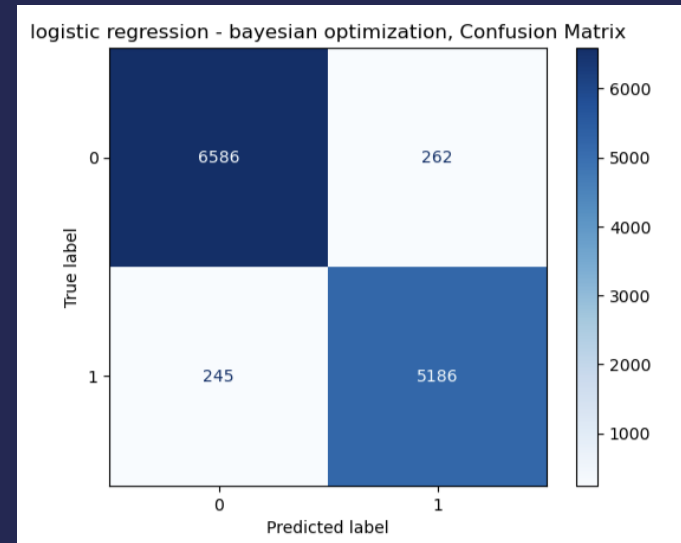
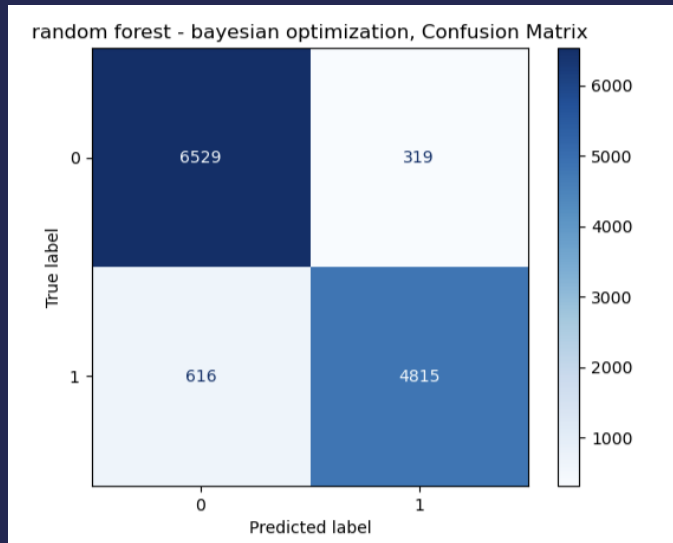
- Machine learning – random forest.
- Machine learning – logistic regression.
- Deep learning – bi-LSTM.
- Transformer – DistilBERT.

Modeling Approach

To avoid data leakage = train/test/split first :

- **ML models:**
 - TFIDF vectorization.
 - Bayesian optimization.
- **Deep learning:**
 - title & text combined.
 - Learning embedding from scratch.
 - Overfitting counter measures.
- **Transformer:**
 - Title & text combined.
 - Tokenization and encoding.

Modeling Approach



Modeling Approach

F1 scores:

- **Random forest = 92 %.**
- **Logistic regression = 96 %.**
- **Deep learning = 97.40 %.**
- **Transformer = 99.50 % = SELECTED.**

Recommendations & Future Work

- Model evaluation on additional datasets = **robustness**.
- “Strong discriminators” & **additional** feature engineering.
- Slight target class **imbalance** & “class_weight” parameter.